EARLY DEMENTIA DETECTION

via Consumer Sensors, Machine Learning & AI Agents

State of the Art • Completed Research • Ongoing Studies • Available Products

GestureCare Research Initiative | February 2026

SENSOR SCOPE OF THIS REPORT

This report focuses exclusively on sensors that most people already own: smartphones (GPS, accelerometer, microphone, screen interaction telemetry), wearables (smartwatch/fitness band with heart rate, HRV, sleep tracking), and home voice assistants (Amazon Alexa, Apple Siri/HomePod, Google Assistant). All findings are evaluated for clinical evidence strength, practical deployability, and readiness for ML/AI pipeline integration.

Executive Summary

Alzheimer's disease and related dementias (ADRD) have a pathological onset 15–20 years before clinical diagnosis. By the time a neurologist sees measurable symptoms, significant neuronal damage has already occurred — a window in which emerging disease-modifying therapies (lecanemab, donanemab) are most effective has likely already closed.

The sensors already in most people's pockets, on their wrists, and in their kitchens generate a continuous, high-resolution behavioral record. That record contains early dementia signal. The field has now matured to the point where this is not a hypothesis — it is established science. The remaining challenges are specificity, generalization across demographics, and the hard problem of labeled longitudinal ground truth.

> *Key finding: Three signal families — gait, sleep architecture, and speech/language — have Tier 1 clinical evidence (multiple independent longitudinal cohorts, AUCs 0.75–0.92, mechanistic links to neuropathology). Consumer hardware captures all three. The field is ready for productized monitoring systems.*

1. Sensor Landscape & Signal Inventory

The three consumer sensor categories below cover the signal space available to most users without any new hardware purchase. Each modality contributes orthogonal signal; the highest-performing systems fuse all three.

1.1 Smartphone — Highest Signal Density

The smartphone is the richest ambient sensor platform in consumer hands. Beyond explicit sensor data, the patterns of interaction with the device itself constitute a high-dimensional behavioral fingerprint.

Passive Hardware Sensors

- Accelerometer/gyroscope: Continuous background gait analysis, tremor detection, postural sway, fall risk, activity classification, and detection of psychomotor slowing — all without any user action.
- GPS: Radius of gyration (daily travel range), number of distinct places visited, routine regularity, mobility restriction. Life-space contraction is an established early dementia marker.
- Microphone (on-device processing only): Voice activity detection frequency, call duration, ambient noise classification. Raw audio never needs to leave the device.
- Ambient light sensor: Circadian light exposure, time outdoors, sleep timing proxy via screen-on/off behavior.
- Wi-Fi/Bluetooth scan logs: Social proximity inference (nearby devices), home/away context, sleep timing proxy via router disconnection.

Interaction Metadata as Cognitive Biomarkers

This is an underappreciated signal channel. Device interaction patterns — independent of any content — are measurable cognitive performance proxies:

- Screen unlock frequency, duration, and time-of-day distribution → sleep disturbance, rumination, behavioral activation.
- Notification response latency (time from notification arrival to first tap) → psychomotor speed, a robust early cognitive decline marker.
- Typing speed, error rate, backspace frequency, inter-key latency → fine motor and cognitive processing speed. Detectable via keyboard telemetry with no content access.
- Charging behavior, alarm setting, and snooze patterns → routine consistency and self-regulation.
- App usage category shifts (less social, more passive entertainment) → apathy, withdrawal.

1.2 Wearables — Best Physiological Signal

Smartwatches and fitness rings (Apple Watch, Garmin, Oura Ring, Fitbit) provide the only continuous physiological signal in the consumer stack. Critical signals include:

Cardiovascular

- Heart rate (HR): Resting HR elevation as autonomic dysfunction proxy.
- Heart rate variability (HRV, RMSSD): Reduced HRV is an established marker of both anxiety/depression and early Alzheimer's autonomic dysfunction. Consumer devices now measure this continuously.

- SpO2 (select devices): Sleep-disordered breathing screening, hypoxia-related cognitive impacts.

Sleep Architecture

This is the highest-value wearable signal for dementia detection. Key metrics:

- Total sleep time, sleep efficiency, wake-after-sleep-onset (WASO).
- Sleep staging estimates (SWS and REM percentages): Reduced slow-wave sleep is mechanistically linked to amyloid-beta clearance failure via the glymphatic system.
- Sleep regularity index (SRI) and day-to-day variability: Recent evidence suggests variability may be an earlier and more sensitive marker than average sleep quality.
- Circadian metrics: Interdaily stability (IS), intradaily variability (IV), circadian amplitude.

Motion & Gait Proxy

- Daily step count and active minutes: Activity reduction is a depression/apathy marker.
- Continuous gait metrics from wrist IMU: Less precise than instrumented walkways but sufficient for longitudinal change detection.
- Restlessness index: Agitation detection, relevant to BPSD in later-stage dementia.

1.3 Smart Speakers — Unique Language Channel

Amazon Echo/Alexa, Apple HomePod/Siri, and Google Nest/Assistant occupy a unique position: they provide a naturalistic, longitudinal speech sample without requiring any structured test protocol. This is the holy grail for speech biomarker research.

Interaction Metadata (No Audio Retention Required)

- Wake word invocation frequency and time-of-day → activity level, sleep timing, routine consistency.
- Silence duration following wake word to command issuance → processing speed metric.
- Command execution success rate; frequency of re-prompting and rephrasing → cognitive-linguistic performance.
- Timer/reminder setting frequency and content → prospective memory and executive function proxy.
- Time-of-night interactions → sleep disturbance indicator.
- Query complexity and vocabulary diversity over time → direct lexical richness metric, a validated dementia marker.

On-Device Audio Features (Privacy-Preserving)

- Pause frequency and duration within speech → established early AD biomarker; extractable without transcription.
- Speech rate (syllables/second), articulation rate, fundamental frequency variability.

- Voice quality metrics: jitter, shimmer, harmonics-to-noise ratio → detectable with no content processing.
- Vocal energy and prosody metrics → depression/anxiety/flat affect proxy.

2. State of the Art — What the Science Supports

The following summarizes the current scientific consensus on what consumer sensors can and cannot reliably detect, organized by signal family and evidence tier.

2.1 Gait Analysis — Tier 1 Evidence

> *Clinical evidence: Gait slowing and increased variability precede cognitive diagnosis by 3–7 years. Dual-task gait performance is detectable in Subjective Cognitive Decline (SCD) before any clinical MCI criteria are met. AUCs 0.75–0.86 in longitudinal cohorts.*

Gait represents the most mature signal in this space. The mechanistic explanation is well-established: prefrontal-cerebellar circuits governing executive function and gait control overlap substantially, and early tau pathology in these circuits produces measurable gait changes years before cognitive symptoms appear.

What is now validated: smartphone IMU gait assessment pipelines producing clinically meaningful metrics — stride velocity, stride time variability (CV), cadence, step length asymmetry, and dual-task cost — from background phone accelerometry during natural walking episodes. The Su et al. (2021) validation against instrumented walkways established that smartphone-derived gait metrics can achieve ICC > 0.85 versus GAITRite gold standard.

Current frontier: passive continuous gait monitoring without any structured test protocol, extracting features from incidental walking detected in background accelerometry. This is now technically feasible and has been demonstrated in research settings. The Seifallahi et al. (2024) Kinect study achieved 85.5% MCI classification accuracy. Wearable-based dual-task gait achieves 86% accuracy (HC vs. MCI/AD) — comparable to paper MoCA testing.

2.2 Sleep Architecture — Tier 1 Evidence

> *Clinical evidence: Reduced SWS and REM sleep, increased fragmentation, and sleep-disordered breathing each independently predict dementia onset 5–10 years before diagnosis. The glymphatic hypothesis provides mechanistic grounding: SWS drives amyloid-beta and tau clearance. Lucey et al. (2021) linked single-night PSG metrics to CSF tau biomarkers in preclinical AD.*

Sleep is the most immediately scalable detection signal because the hardware already exists in millions of consumer wrists. The Heremans et al. (2025) study achieved 0.90 classification accuracy for AD using simultaneous PSG + single-channel EEG wearable, with the key finding that a wristband-level device is sufficient when combined with AI sleep staging (SeqSleepNet).

Day-to-day sleep variability — not just mean sleep quality — is emerging as an earlier and more sensitive marker. Baril et al. (2024) linked variability in sleep timing and duration to Alzheimer's biomarkers, suggesting that erratic sleep patterns may signal underlying pathology before average sleep quality degrades.

Key limitation: Existing consumer wearable sleep staging validations are predominantly in healthy young adults. Performance degrades in cognitively impaired populations, and the 15% staging error in consumer devices vs. PSG remains a meaningful accuracy gap that requires correction in any serious monitoring pipeline.

2.3 Speech & Language — Tier 1 Evidence

> *Clinical evidence: Fraser et al. (2016) DementiaBank: 35 linguistic features achieve 81.9% classification of AD vs. control from narrative speech. Multiple subsequent studies: AUCs 0.80–0.92. Eyigoz et al. (2020) predicted AD onset years in advance from speech recorded before clinical diagnosis.*

Language impairment is among the earliest cognitive changes in Alzheimer's disease, reflecting pathology in temporal and frontal language networks. The field has progressed from simple word-count metrics to multi-domain feature extraction spanning acoustic, lexical, syntactic, and semantic dimensions.

Critical finding for consumer deployment: Acoustic features alone — extractable without any transcription — show substantial discriminative ability. Ding et al. (2024) Framingham Heart Study demonstrated MCI classification from acoustic voice features only, with no semantic content processing required. This enables fully privacy-preserving on-device deployment.

Lampert et al. (2025) JMIR tested cross-linguistic generalization across English and Spanish speakers (N=211), finding that some acoustic features generalize cross-linguistically while others do not — a critical finding for multilingual deployment.

Smart speaker opportunity: Natural conversational speech with Alexa or Siri represents a continuous, passive longitudinal speech sample. Passive speech monitoring via triggered feature extraction during natural device interactions — without any structured test protocol — is the logical next deployment step and is now technically feasible.

2.4 Circadian Rhythm — Tier 1 Evidence

Circadian disruption precedes clinical dementia by up to a decade and is mechanistically linked to amyloid and tau pathology through glymphatic clearance and clock gene expression. The actigraphy-based rest-activity rhythm literature is substantial, with Interdaily Stability (IS) and Intradaily Variability (IV) independently associated with dementia risk across multiple cohorts.

Consumer hardware can now derive meaningful circadian metrics: IS, IV, mesor, amplitude, and acrophase from smartphone accelerometry and wearable activity data. The challenge is separating biologically-driven circadian deterioration from lifestyle-driven disruption (shift work, caregiving, social schedules).

2.5 Fine Motor & Device Interaction Latency — Tier 2 / Novel

Psychomotor slowing is one of the most robust and earliest features of cognitive decline. Device interaction latency — time from notification arrival to first response, wake-word-to-speech-onset on smart speakers — constitutes a continuous, ecologically valid psychomotor speed measurement.

Keyboard dynamics (inter-key timing, error rate, backspace frequency) have demonstrated MCI classification AUCs of 0.70–0.80 in research settings. The mPower Parkinson's study validated this model at scale for neurodegenerative disease. Touchscreen typing dynamics from naturalistic smartphone use are now considered a viable passive cognitive performance metric.

> *Novel hypothesis with high potential: Smart speaker response latency (wake-word to utterance onset) and command success rate have not yet been validated as cognitive biomarkers in a prospective study, but the mechanistic case is strong. This represents an unclaimed evidence niche with high practical impact given the hardware is already deployed at massive scale.*

2.6 Signal Evidence Summary

Signal	Consumer Device	Evidence Tier	Best Reported AUC	Key Limitation
Gait speed & variability	Phone IMU	Tier 1	0.85–0.88	Dual-task needs prompting
Sleep SWS/REM/fragmentation	Smartwatch/ring	Tier 1	0.90 (EEG+accel)	Consumer staging ±15% vs PSG
Speech: pause, rate, lexical richness	Phone/Smart speaker	Tier 1	0.81–0.92	Labeled longitudinal data scarce
Circadian rhythm (IS/IV)	Phone/Wearable	Tier 1	Cohort-based	Lifestyle confounders hard to separate
Fine motor / typing dynamics	Smartphone	Tier 2	0.70–0.80	Parkinson confound
Mobility / GPS radius	Phone GPS	Tier 2	Moderate	Depression confounders; rural/urban bias

Signal	Consumer Device	Evidence Tier	Best Reported AUC	Key Limitation
HRV / autonomic function	Smartwatch	Tier 2	Moderate	Weak specificity to dementia vs depression
Device interaction latency	Phone/Smart speaker	Novel (Tier N)	Unvalidated	No prospective study yet
Smart speaker query complexity	Alexa/Siri/Google	Novel (Tier N)	Unvalidated	Requires longitudinal instrumentation

3. Completed Research — Key Studies

The following represent landmark published studies that established the evidence base for consumer sensor dementia detection. This is a curated selection; the full citation list in the companion AI_MentalHealth_Citations.docx encompasses 71+ primary sources covering 200+ underlying studies.

3.1 Smartphone Passive Sensing

Stamatis et al. (2024) — npj Mental Health Research

N=1,013 — one of the largest smartphone passive sensing studies. GPS, app use, and communication data over 16 weeks correlated with PHQ-8 and GAD-7 outcomes. Key finding: different features predict depression vs. anxiety at different time lags. Distal features (2-week window) differ from proximal (concurrent). This established that sensor-symptom relationships are temporally complex and require multi-scale modeling.

Walsh et al. (2025) — Psychiatry Research

Reviews 112 papers across digital phenotyping. GPS and accelerometer used in 91/112 studies. Critical finding: massive heterogeneity in feature extraction methods prevents cross-study validation. Essential reading before designing any new feature pipeline — identifies the methodological fragmentation problem that is the field's most pressing infrastructure gap.

Guo et al. (2025) — JMIR

Scoping review of 42 studies across depression, anxiety, bipolar, and schizophrenia. CNN-LSTM achieved 92.16% accuracy for anxiety. Depression best predicted by HR and step count; anxiety by sleep and social interaction; schizophrenia by movement index. Only 2% of studies had external validation — a stark indicator of the generalization problem.

3.2 Sleep & Wearables

Heremans et al. (2025) — npj Aging

N=67 cognitively normal elderly + 35 AD patients. Simultaneous PSG + wearable EEG/accelerometry. AI sleep staging (SeqSleepNet). Classification accuracy 0.90 for AD; 0.76 for prodromal AD. Critical finding: single-channel EEG + accelerometry is sufficient — validating low-cost wearable design for clinical screening.

Zuurbier et al. (2021) — Alzheimer's & Dementia

Rotterdam Study cohort, N=1,322, up to 11.2-year follow-up, 60 incident dementia cases. Actigraphy-measured nighttime wakefulness predicted AD onset. One of the strongest longitudinal evidence papers for the sleep-dementia link using objective measurement. Establishes that the signal is present at population scale, not just in clinical samples.

Pinto et al. (2026) — JMIR (49 studies, >200,000 participants)

Most comprehensive wearable review available. Covers actigraphy, consumer wearables, sleep, circadian rhythms, and activity across the dementia detection literature. Key finding: ML/DL is underutilized compared to statistical models — substantial performance headroom remains for AI-driven approaches.

3.3 Speech & Language

Fraser, Meltzer & Rudzicz (2016) — Journal of Alzheimer's Disease

Landmark NLP paper on DementiaBank Pitt Corpus. 35 linguistic features achieve 81.9% classification of AD vs. control from narrative speech. Established the NLP-for-dementia field. Every subsequent speech study cites this. The feature set — pause patterns, lexical richness, syntactic complexity — remains the baseline against which new work is measured.

Eyigoz et al. (2020) — EClinicalMedicine (The Lancet)

Longitudinal speech analysis predicting AD onset years in advance. NLP on connected speech from participants who later developed AD. Key evidence that speech biomarkers are predictive (not just diagnostic) — the signal exists before clinical criteria are met. This is the most important paper for arguing early detection feasibility.

Ding et al. (2024) — JMIR Aging (Framingham Heart Study)

N=200 (100 MCI, 100 matched controls). Speaker diarization + acoustic feature extraction from voice recordings. Privacy-preserving: no transcription required. AUC demonstrated for acoustic-only MCI detection. This is the strongest evidence that a fully on-device, no-content-transcription pipeline can detect MCI from voice alone.

Amini et al. (2024) — Alzheimer's & Dementia (Framingham)

Automated NLP pipeline on neuropsychological interview audio. NLP model AUC 71.3%, matching standard neuropsychological test scores. Combined with APOE and health factors: AUC 71.7%.

Demonstrates that automated interview analysis can match clinical assessments — the bar for consumer deployment validation.

3.4 Multimodal & Ambient

TIHM Study — Palermo et al. (2023)

56 dementia patients' homes, avg 50 days/participant. PIR motion sensors, door sensors, under-mattress sleep sensor, physiological devices. Labels for agitation, UTI, sleep disturbance. The TIHM dataset remains the only large open benchmark for home ambient sensor + dementia research, and is now publicly available on Zenodo (doi: 10.5281/zenodo.7622128).

Iaboni et al. (2022) — Alzheimer's & Dementia DADM

N=31 dementia inpatients. Multisensor wristband (ACC, EDA, skin temperature, PPG). Key finding: personalized models significantly outperformed population-level models for minute-by-minute BPSD detection. This is the definitive evidence that the field must move from population norms to personalized baselines. Sensor patterns are highly individualized.

4. Ongoing Research

The following studies and programs are active as of early 2026. This represents the frontier of the field — studies that will define best practices for the next generation of consumer sensor monitoring systems.

4.1 Active Major Studies

SERENADE Project (EU-Funded)

Castellano et al. (2025, arXiv 2504.08877). EU-funded project collecting 1 year of smart home + wearable data from 30 MCI patients living alone. Explicit focus on explainable AI for clinical trust. In progress. Key ongoing project to monitor for forthcoming primary results — will be the first long-duration naturalistic MCI monitoring dataset with explainability-focused analysis.

PREVENT Dementia Cohort (UK)

Midlife dementia prevention study with detailed cognitive assessment and blood/imaging biomarkers. Passive monitoring expansion is actively in negotiation. This is the highest-value cohort for passive sensor validation because it captures participants 15–20 years before expected clinical presentation — the true early detection window.

DementiaBank Longitudinal Speech Extension

The DementiaBank consortium is actively expanding the Pitt Corpus with longitudinal speech samples and improved biomarker annotation. Current limitation: single time-point recordings from most participants. The longitudinal extension will enable training of trajectory models rather than cross-sectional classifiers.

Su et al. Protocol — JMIR Research Protocols (2024)

North Bristol NHS Trust / Join Dementia Research. Feasibility and acceptability study for remote sleep and circadian rhythm monitoring in MCI/dementia using EEG headband, actigraphy, sleep diary, cognitive tests, and salivary melatonin/cortisol. This protocol represents the most comprehensive multimodal home monitoring design currently active.

4.2 Key Open Research Gaps (Active Investigation Areas)

Consumer Wearable Validation in Older Adults

Existing consumer wearable sleep staging validations are predominantly in healthy young adults aged 18–40. Active work is underway at multiple sites to characterize performance degradation in older adults with comorbidities. This is a critical gap before consumer devices can be used in clinical applications.

Cross-Linguistic Speech Biomarkers

The ADReSS-M challenge (English + Greek) and Lampert et al. (2025) are active efforts to characterize which speech features generalize across languages. Multiple groups are now attempting validation in Spanish, Mandarin, and other high-prevalence languages.

Device Interaction Latency as Biomarker

No published prospective validation exists yet for using smart speaker response latency or smartphone notification response time as cognitive biomarkers. Several groups are now designing instrumentation protocols. This is the highest-potential novel signal — the hardware is already massively deployed, the mechanism is sound (psychomotor slowing), and the first group to validate it will establish prior art for a high-value clinical tool.

Personalized Baseline Methods for Longitudinal Change

The field is shifting from population-norm comparison to within-person change detection. Active methodological work on Bayesian change-point detection, CUSUM, and Gaussian process regression applied to multi-sensor longitudinal streams. Personalized models outperform population models for BPSD detection (Iaboni 2022) and for schizophrenia relapse prediction (Lamichhane 2023).

4.3 Federated Learning Infrastructure

Multiple groups are developing federated learning frameworks for privacy-preserving model training across devices and sites. Key challenge: heterogeneous hardware (different wearable brands produce incompatible feature distributions) makes federated training technically difficult. Active research into hardware-agnostic feature normalization and modality-dropout training for missing-data resilience.

5. Products Currently Available

The market divides into three categories: clinical-grade monitoring systems, consumer wellness products with research-validated signals, and platform infrastructure providers enabling third-party development.

5.1 Clinical-Grade Monitoring Systems

Product / Company	Sensor Focus	Clinical Status	Notes
Altoida (Altoida Inc.)	Smartphone: AR task, gait, fine motor, eye tracking	FDA Breakthrough Device Designation	Cognitive assessment platform. ~20 min smartphone-based test. Generates 'Neurological Performance Index'. Not continuous passive monitoring.
Winterlight Labs	Speech analysis from tablet recordings	Research / clinical trials	Automated Cookie Theft picture description analysis. AUC ~0.87 for MCI. Licensing to pharma for trial endpoints.
Linus Health (DCTclock)	Digital clock drawing + cognitive tasks	CE Mark; FDA 510(k) cleared (2023)	Quantifies 500+ features from clock drawing. Used in primary care screening. Not passive monitoring.
Cognetivity Integrated Cognitive Assessment (ICA)	Smartphone + tablet cognitive tasks	CE Mark; Health Canada approved	Cat/non-cat image classification task. Targets early Alzheimer's. 5-minute test.
BrainCheck	iPad-based cognitive battery	FDA 510(k) cleared	7 games adapted from validated neuropsychological tests. Point-of-care use.
ActiGraph (Theralink)	Research-grade actigraphy	IRB / research use; FDA cleared for prescription use	Gold standard actigraphy. Widely used in ADNI, UK Biobank. Not consumer.

5.2 Consumer Wearables with Research-Validated Signals

These are not marketed as medical devices but have published validation data for signals relevant to dementia detection:

Device	Key Validated Signals	Research Use	Limitation for Dementia Detection
Oura Ring (Gen 3/4)	Sleep staging (SWS, REM, efficiency), HRV, body temperature, SpO2	Used in 20+ published studies including Brooks et al. (2025)	Sleep staging accuracy ~86–90% vs PSG in young adults; less validated in elderly
Apple Watch (Series 9 / Ultra 2)	HRV, sleep staging, gait (ataxia screening), AFib detection, fall detection	Used in Apple Heart Study (419k participants), FDA-cleared for AFib	Sleep staging less accurate than Oura; gait metrics not yet validated for MCI
Garmin (Fenix, Forerunner)	Sleep staging, HRV, Body Battery, Pulse Ox	Several published validation studies	Proprietary algorithms limit research reproducibility
Fitbit (Pixel Watch)	Sleep staging, SpO2, HRV, EDA (Sense 2)	Google's large-scale health datasets	EDA implementation limited; Google's research roadmap opaque post-acquisition
Withings ScanWatch 2	ECG, SpO2, sleep staging, breathing disturbances	Multiple EU clinical studies; FDA cleared ECG	Less gait/motion analysis than Apple Watch

5.3 Smart Speaker Integration — Current State

As of early 2026, no consumer smart speaker product explicitly positions itself as a cognitive monitoring tool. However, the infrastructure is partially in place:

Amazon Alexa / Echo

- Alexa Together: Remote monitoring subscription for older adults. Currently limited to activity detection (motion sensors, routine check-ins, button presses) and fall detection alerts. No speech biomarker analysis.
- Alexa Health Skills Kit: Allows third-party skills to implement health assessments. Winterlight Labs has implemented a speech assessment skill using this API.
- Key gap: No longitudinal acoustic feature extraction from natural Alexa interactions. This is the highest-value near-term opportunity.

Apple Siri / HomePod

- Apple Health integration allows third-party apps to write cognitive assessment results to Health app. Apple's on-device AI processing (Private Cloud Compute) aligns well with privacy-preserving speech feature extraction.

- No current passive monitoring product. Apple's roadmap suggests Health app expansion, but no confirmed cognitive monitoring feature.

Google Assistant / Nest Hub

- Google's Nest Hub has a sleep sensing radar (Soli chip) that detects sleep staging without contact. Google Health Studies app has conducted large-scale passive monitoring research.
- No current commercialized dementia monitoring product. Research partnerships with UCSF and other academic medical centers active.

5.4 Platform & Infrastructure Providers

Platform	Function	Relevance
mindLAMP (Beth Israel / Harvard)	Open-source digital phenotyping platform for smartphones. GPS, accelerometer, surveys, cognitive tasks.	Gold standard research deployment platform. Used in dozens of published studies. Free.
AWARE Framework	Open-source passive sensing on Android/iOS. Accelerometer, GPS, Bluetooth, app use, screen events.	Research standard for smartphone sensing studies. Supports custom sensor plugins.
Apple Research Kit / ResearchKit	iOS framework for research studies. Cognitive task library, passive sensing hooks.	Used in major consumer health studies. Requires institutional partnership.
Amazon HealthLake	FHIR-compliant data lake for health data. ML ready.	Infrastructure layer for aggregating wearable + EHR data. Not end-user product.
GestureCare (this initiative)	FHIR-based care coordination + sensor fusion platform targeting elder care.	Defensive publication establishes prior art across full sensor and inference design space.

6. Machine Learning & AI Architecture for Detection

This section outlines the current best-practice ML/AI pipeline architecture for a consumer sensor dementia monitoring system, based on the published literature and the two-stage pipeline described in the GestureCare research documents.

6.1 The Two-Stage Pipeline

> *Stage 1 — State Inference (Proximal Signals): Infer current affect, arousal, stress load, agitation, apathy, engagement, and social withdrawal from sensor data. This is the moment-to-moment classifier.Stage 2 — Change Detection (Distal Clinical Meaning): Detect departures from that person's personal baseline over weeks to months. This is the early warning system. The 'baseline + deviation' framing is essential because most signals are person-specific and drift with age, seasonality, medications, and comorbidities.*

6.2 Model Architecture Recommendations

Phase 1: Unimodal Baselines (Months 0–12)

Before fusion, establish rigorous unimodal baselines for each Tier 1 signal independently. This establishes the incremental value of fusion and produces clinically interpretable single-signal models deployable earlier than complex fusion architectures.

- Simple, interpretable architectures first: logistic regression, random forest, gradient boosting on hand-crafted features before adding deep learning complexity.
- Target AUC > 0.75, sensitivity > 0.70 at 90% specificity for each unimodal classifier.
- Site-stratified or participant-stratified cross-validation is essential — avoiding data leakage is the most common error in this literature.

Phase 2: Multimodal Fusion (Months 6–24)

- Late fusion: Independent per-modality models combined via weighted ensemble. More robust to missing modalities in deployment.
- Attention-based fusion: Transformer encoder over multi-modal feature sequences, allowing dynamic modality weighting based on data quality and context.
- Missing modality handling: Design fusion architectures that degrade gracefully when one sensor stream is unavailable. Modality dropout during training is a proven technique.
- Target AUC > 0.85; statistically significant improvement over best single modality.

Phase 3: Longitudinal Trajectory Modeling (Months 12–36)

- Shift from cross-sectional classification (MCI vs. normal at a point in time) to longitudinal trajectory modeling (predicting conversion from current trajectory).
- Bayesian change-point detection on latent state scores: CUSUM, EWMA, Page-Hinkley test.
- Gaussian process regression over multi-modal feature time series for continuous cognitive trajectory estimation.
- Uncertainty quantification is non-negotiable for clinical deployment: calibrated confidence intervals via conformal prediction or Bayesian deep learning.

6.3 LLM and Agent Integration

The most recent development (2024–2025) is the integration of large language models not as classifiers but as reasoning and synthesis layers:

LLM as Clinical Synthesizer

- FHIR-to-LLM translation: Convert structured sensor data, observation timelines, and drift metrics into contextually rich prompts for clinical reasoning.
- LLMs can identify complex patterns across multiple time series that do not reduce to simple statistical anomalies — e.g., 'patient is sleeping less AND typing more slowly AND using shorter sentences in Alexa queries over the past 6 weeks.'
- Natural language report generation: Synthesize multi-modal sensor trends into clinician-readable summaries and family-accessible explanations.

Agent Architecture for Monitoring

- Sensor agent: Continuously ingests raw sensor streams, applies feature extraction, quality flagging, and artifact removal.
- State inference agent: Runs Stage 1 classifiers; maintains current-state estimates for each monitored condition.
- Change detection agent: Runs Stage 2 longitudinal models; tracks personal baselines and deviation scores.
- Clinical synthesis agent (LLM): Receives outputs from all agents, applies contextual reasoning, generates alerts and explanations.
- Human-in-the-loop: All alerts go to clinician or caregiver review before any clinical action. The system informs; humans decide.

6.4 Context Disambiguation — The False Alarm Problem

> *The most common failure mode: 'stress score' from consumer wearables is non-specific. Exercise, excitement, pain, fever, and anxiety can produce identical HR/HRV/EDA signatures. A key research direction is context disambiguation — what was the person doing? — rather than better physiological signal alone.*

Required disambiguation layers:

- Activity classification: sleep / exercise / social outing / rest before interpreting any physiological signal.
- Illness detection: fever elevates HR; respiratory infection changes voice quality. Temperature + HR + respiration rate together can flag acute illness vs. baseline drift.
- Medication timing: benzodiazepines, anticholinergics, antipsychotics all affect HRV, sleep staging, gait, and speech. Medication timing logs are essential covariates.
- Seasonal/environmental: circadian metrics need weather and light exposure correction.

7. Key Open Datasets for Development

Dataset	Content	Access	Best Use
DementiaBank Pitt Corpus	Picture description speech (Cookie Theft), narrative recall, AD + control, neuropsychological scores	Application required (dementia.talkbank.org)	Speech biomarker development, NLP classifier training
TIHM Dataset (Zenodo)	PIR, door, sleep, physiological data from 56 dementia patient homes. 2,803 person-days. Labeled for agitation, UTI, sleep disturbance.	CC license, direct download (Zenodo 10.5281/zenodo.7622128)	Home ambient sensor + dementia monitoring benchmarking
ADReSS / ADReSS-M Challenge Data	DementiaBank derivative. English + Greek multilingual AD speech. Challenge results from multiple teams.	Via DementiaBank access	Multilingual speech biomarker validation, benchmark comparison
CrossCheck Dataset	63 schizophrenia patients, ≤1 year monitoring, 6 passive sensing modalities (smartphone)	Contact Ben-Zeev group	Smartphone passive sensing for psychiatric relapse prediction
UK Biobank Accelerometry	Wrist accelerometry from 100,000+ participants, linked health records	UK Biobank access application	Circadian and sleep metrics at population scale; limited cognitive phenotyping
ADNI (Alzheimer's Neuroimaging Initiative)	Longitudinal cognitive, imaging, CSF, blood biomarkers. No passive sensors but gold-standard labels.	adni.loni.usc.edu	Model validation against gold-standard cognitive and biomarker outcomes
PhysioNet	Physiological signal databases, some dementia-relevant	Open access (physionet.org)	HRV, ECG, actigraphy baseline datasets

8. Known Hard Problems — Honest Assessment

Any serious research or product effort in this space must grapple with the following unsolved problems. These are not theoretical limitations — they are demonstrated failure modes in published literature.

8.1 The Ground Truth Problem

The single largest barrier to progress is labeled training data. Existing longitudinal dementia cohorts (ADNI, UK Biobank, AIBL) were not designed for passive sensor data collection. They have gold-standard cognitive and biomarker assessments but minimal continuously collected behavioral data. Conversely, large passive sensor datasets lack deep clinical phenotyping. Bridging this gap is the field's most urgent infrastructure need.

Minimum viable cohort for training a longitudinal dementia detection model: N > 300 with MCI at enrollment, annual neuropsychological assessment, 18-month minimum follow-up, 20–30% expected conversion rate, biomarker confirmation in a subsample (minimum N=100 amyloid PET or CSF).

8.2 Specificity — The Non-Specific Signal Problem

Reduced activity ≠ depression (could be pain, weather, caregiving burden). Elevated HR/low HRV ≠ anxiety (could be exercise, fever, excitement). Sleep fragmentation ≠ dementia (could be sleep apnea, nocturia, pain). Every signal in this space is multiply determined. Context disambiguation and multi-modal corroboration are not optional enhancements — they are the minimum bar for clinical utility.

8.3 Demographic Generalization

Models trained on young adults fail in older adults. Models trained in North American English fail in Spanish. Models trained on predominantly white populations show degraded performance in populations with different skin tones (affecting PPG accuracy), different speech patterns, and different cultural behavioral norms. Every paper in this space trained on a narrow demographic should be treated as pilot evidence only.

8.4 Consumer Wearable Accuracy vs. Clinical Grade

Consumer wearable sleep staging has ~15% error vs PSG gold standard, predominantly in older adults with comorbidities — exactly the population of interest. HRV from PPG has error vs ECG that compounds across the feature extraction chain. These are not disqualifying limitations, but they require correction models and cannot be ignored in validation claims.

8.5 Causal Direction & Confounders

Longitudinal observational studies establish association, not causation. Sleep disruption, reduced activity, and social withdrawal could be early dementia symptoms or the causes of cognitive decline. This matters for intervention design. Medication timing, acute illness, seasonal effects, major life events, and socioeconomic factors all drive behavioral changes that can be misclassified as dementia prodrome.

9. Recommended Development Priorities

Based on the evidence review, the following priority ordering maximizes the probability of producing a clinically credible, deployable consumer sensor monitoring system within a 24–36 month horizon:

Priority 1: Build the Longitudinal Data Infrastructure First

Partner with a memory clinic or aging cohort to deploy passive monitoring alongside gold-standard neuropsychological assessments. No amount of algorithmic sophistication compensates for inadequate labeled training data. Without longitudinal ground truth, any model is validatable only in cross-sectional settings — the wrong test for the clinical question being asked.

Priority 2: Implement Tier 1 Unimodal Baselines

Gait (from phone IMU), sleep (from wearable), and speech (acoustic features from smart speaker or phone) are the three signals with the strongest evidence. Implement clean, reproducible feature extraction pipelines for each. Validate against published benchmarks (DementiaBank, TIHM). Establish that your pipeline reproduces published AUCs before attempting fusion.

Priority 3: Personalized Baseline, Not Population Norm

The Iaboni et al. (2022) finding — that personalized models significantly outperform population-level models for BPSD detection — is probably the most important architectural decision in this literature. Design for within-person change detection from the start. A 30-day enrollment period to establish individual baseline is the minimum; longer is better.

Priority 4: Context Disambiguation Before Signal Fusion

Activity classification (sleep/exercise/rest/social) must gate all physiological inference. A stress spike during a morning jog is not an anxiety episode. Illness detection (fever proxy) must suppress sleep and HR anomaly alerts. Medication timing must be a covariate in all models. Build these disambiguation layers before adding more sensor modalities.

Priority 5: Validate Smart Speaker Response Latency

This is the highest-potential unclaimed niche in the consumer sensor space. Design a prospective study to instrument natural Alexa/Siri interactions with millisecond-precision response latency logging. Pre-register the study before collection. First validated paper on this signal will establish prior art for a high-value biomarker that no current product captures.

10. Conclusion

The science is ahead of the products. The evidence that consumer sensors capture early dementia signal — in gait, sleep, speech, and behavioral patterns — is no longer a hypothesis; it is established across multiple independent longitudinal cohorts. The technical infrastructure for ML/AI analysis of these signals has matured substantially, and the hardware is already in the homes of the people who need monitoring.

The gap is in three places: labeled longitudinal training data at scale; validated demographic generalization; and the clinical workflow integration that makes AI-generated alerts actionable rather than merely interesting. The first group to close all three gaps with a consumer-accessible, privacy-preserving, personalized monitoring system will have produced something with genuine clinical impact — the ability to shift dementia diagnosis 5–10 years earlier, into the window where emerging disease-modifying therapies may actually work.